

Chapter 43: Hemorrhagic Placental Disorders —

Revision Notes

Williams Obstetrics 26e, pp. 1943–1999 of the PDF. High-yield numbers are in **bold**. The four tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 14 figures are included from the PDF (Fig 43-11 is printed three times, on pp. 1975–1977; one copy is used).

Big picture: At term **>600 mL/min** flows through the intervillous space, so placental separation can be torrential. Three placental disorders contribute substantially to maternal deaths in the second half of pregnancy: **placental abruption**, **placenta previa** and **placenta accreta spectrum (PAS)**. PAS is now more common because of rising cesarean rates.

1. Placental Abruption

Definition and etiopathogenesis

- **Placental abruption (abruptio placentae)** = partial or total separation of the placenta from its implantation site **before delivery** ("rending asunder"; usually a sudden event).
 - "Premature separation of the normally implanted placenta" is the most exact term because it **excludes previa**.
- Likely starts with **rupture of a decidual spiral artery** → bleeding into the **decidua basalis** → expanding **retroplacental hematoma** splits the decidua (a thin layer stays on the myometrium) and lifts and compresses the placenta.
- Predispositions: **impaired trophoblast invasion with atherosclerosis** (preeclampsia), inflammation or infection. Histology **cannot date** the abruption.
- Early abruption may be asymptomatic.

External vs concealed hemorrhage

- Usually blood dissects between membranes and uterine wall and escapes through the cervix → **external bleeding**.
- Less often blood is retained → **concealed hemorrhage** → delayed diagnosis, **greater maternal and fetal hazard**, and **more consumptive coagulopathy** (higher intervillous pressure forces more **placental thromboplastin** into the maternal circulation).

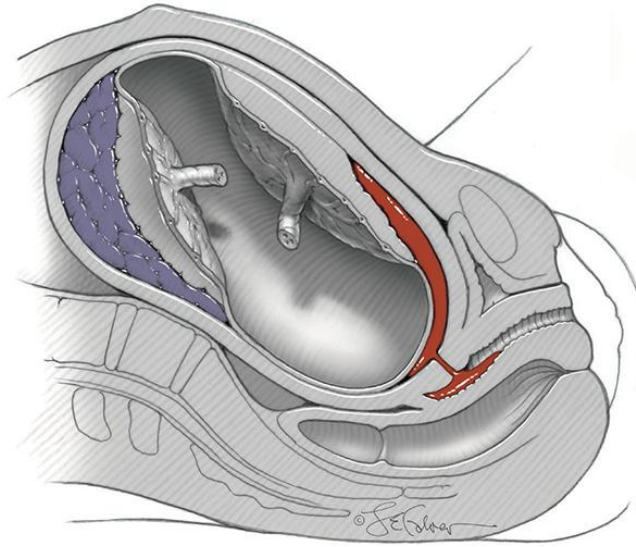


Figure 43-1. Left: total abruption with concealed hemorrhage. Right: partial abruption with blood dissecting between membranes and decidua to the cervix (external bleeding).

Source of blood: maternal

- Nontraumatic abruption: blood is **almost all maternal** (villi are initially intact).
 - Fetal-to-maternal hemorrhage in only **20%**, all **<10 mL** (78 women, Parkland). Fetal cells in maternal blood in only **4%** (68 women).
- Placenta: a **circumscribed depression** a few cm wide covered by **dark clot**. A very fresh separation may look **normal** (changes take minutes). **Clot "age" cannot be determined** exactly.

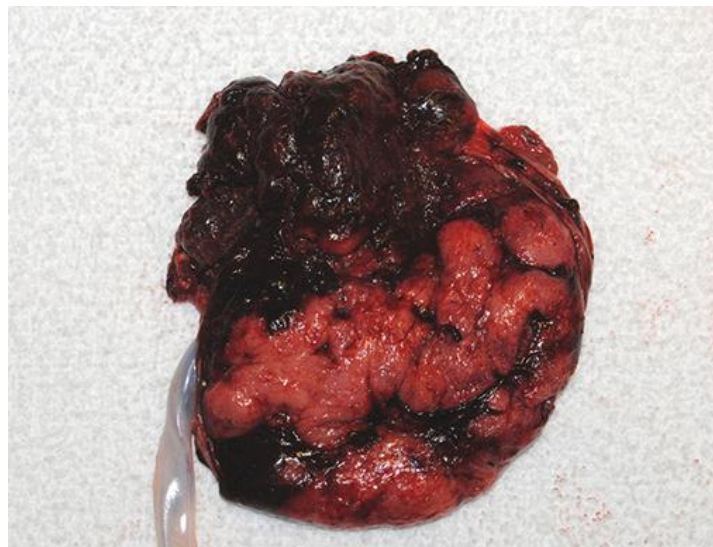


Figure 43-2. Partial abruption: a dark, adherent clot that has depressed the placental substance (probably several hours old).

Severity

- Williams: abruption is **severe when the fetus dies**.
- **Ananth definition of severe abruption** (≥ 1 of):
 1. **Maternal**: DIC, shock, transfusion, hysterectomy, renal failure, death
 2. **Fetal**: nonreassuring status, FGR, death
 3. **Neonatal**: death, preterm delivery, FGR

Traumatic abruption

- Motor vehicle crashes, assault; can follow **relatively minor trauma**.
- **Fetomaternal hemorrhage is more common** than in spontaneous abruption (placental tears/"fractures"): **80–100 mL in 3 of 8** Parkland cases.

- A nonreassuring FHR (e.g., **sinusoidal**) may be the **only sign** of separation.
- ⚠ After blunt abdominal trauma, **uterine contractions are the single most important predictor** of abruption (see Ch 50).

Chronic abruption

- Separation **not followed by delivery**; may start early in pregnancy.
- Associations: abnormally ↑ **maternal serum aneuploidy markers**; 1st- and 2nd-trimester bleeding → 3rd-trimester abruption.
- **CAOS = chronic abruption–oligohydramnios sequence.**
- Arrested late hematomas may show ↑ **MSAFP** or placenta-specific RNAs.

Frequency

- **~0.5% (1 in 200)** deliveries on average; **nearly 1%** in a US database of >125 million births; **1 in 290** at Parkland (2000–2015).
- Rising in the US, mostly in **Black women** (partly conservative management of early preeclampsia).
- **Abruption severe enough to kill the fetus** at Parkland: **0.24% (1 in 420)** in 1956–1967 → **0.05% (1 in 2060)** through 2020 (fewer high-parity births; more prenatal care and emergency transport).

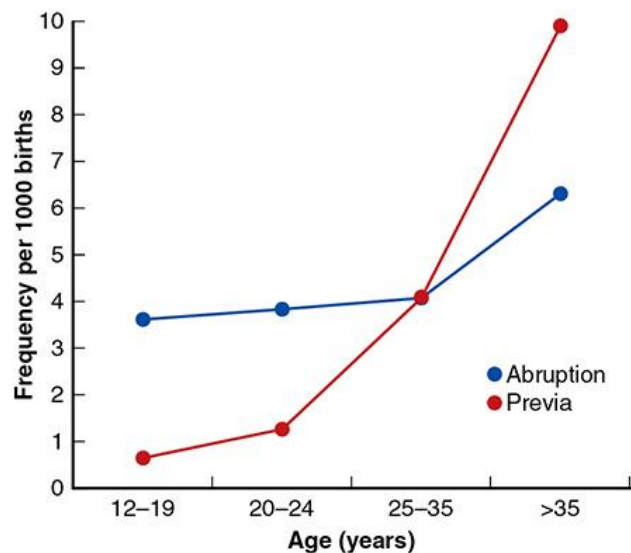


Figure 43-3. Abruptio (~4 per 1000) rises modestly after 35 yr (~6.3); previa climbs steeply with age (~0.6 → ~10 per 1000 after 35 yr).

Perinatal morbidity and mortality

- Frequency rises across the 3rd trimester: **more than half** of Parkland abruptions occurred at **≥37 wk**. Morbidity and mortality are worse with **earlier abruption** and with **preeclampsia**. Associated with **major congenital anomalies**.
- **10–12% of all 3rd-trimester stillbirths** at Parkland are from abruption.
- **PMR 119/1000** with abruption vs **8/1000** overall (US, >15 million births).
- **15% of liveborn neonates die** (Parkland). Deaths mainly from **asphyxia and prematurity**.
- 20% had severe acidemia and **20% of survivors had cerebral palsy** (43 liveborns); **35%** had umbilical artery **pH <7.00** (84 liveborns).

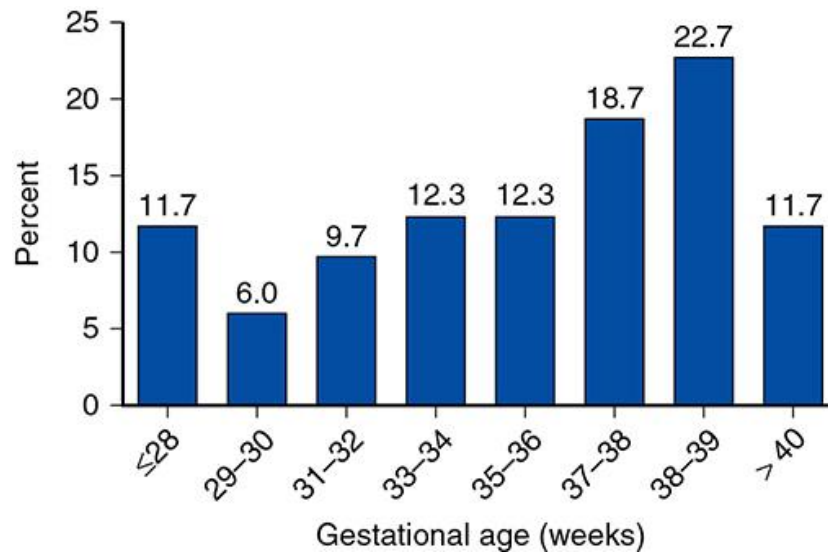


Figure 43-4. Abruption by gestational age at Parkland: >50% at ≥ 37 wk. (The x-axis label printed "38–39" after "37–38" is probably 39–40.)

Predisposing factors

Table 43-1. Risk Factors for Placental Abruption

Risk Factor	Relative Risk
Prior abruption	10–188
Increased age and parity	1.3–2.3
Preeclampsia	2.1–4.0
Chronic hypertension	1.8–3.0
Chorioamnionitis	3.0
Preterm ruptured membranes	2.4–4.9
Multifetal gestation	2–8
Low birthweight	14.0
Hydramnios	1.9
Cigarette smoking	1.4–1.9
Single umbilical artery	3.4
Cocaine use	—
Uterine leiomyoma	2.6
Subchorionic hematoma	5.7

Data from Aviram, 2015; Downes, 2017; Gutvitz, 2016; Khazaei, 2019; Kortekaas, 2020; Morgan, 2016; Ruiter, 2015; Tuuli, 2011.

Demographic

- **Advancing maternal age** (data on great parity conflict).
- Race: abruption killing the fetus in **1 in 200** Black and white women, **1 in 300** Asian, **1 in 350** Latina (366,000 Parkland deliveries).
- **Familial**: a sister's risk is **doubled** after a severe abruption. Candidate genes in mitochondrial pathways.

Hypertension: the most frequent association

- Gestational hypertension, preeclampsia, chronic hypertension; **HELLP** ↑ risk. Expectant management of preterm preeclampsia → abruption in **4%**.

- 408 abruptions with fetal death: **hypertension in half** once hypovolemia was corrected; **¼ of all** had chronic hypertension.
- Chronic hypertension: abruption **1.5%** (MFMU). Treated chronic hypertension **with significant proteinuria ~1%** vs **0.3%** without. Superimposed preeclampsia or FGR → higher still.
- **Severity of hypertension does not correlate** with abruption incidence; but preeclampsia + abruption → worse outcomes.
- Prior abruption → ↑ long-term **cardiovascular mortality**.
- **MgSO₄** (Magpie) may ↓ abruption; **aspirin 100 mg/d** may ↓ abruption (meta-analysis).

Preterm prelabor rupture of membranes

- **5%** with ruptured membranes at 20–36 wk; **17%** with **previable** PPROM; higher still with **infection**.
- **Hydramnios: 8x** at ≥34 wk (one cohort); **2x** in a meta-analysis. **Abrupt uterine decompression** may trigger it.

Prior abruption (recurrence)

Study	Recurrence
Abruption with fetal death (Pritchard)	12% , half with another fetal death
Furuhashi	22% ; half recurred 1–3 wk earlier
Dutch longitudinal (Ruiter)	5.8%
Population (767,000)	7x for "mild", 12x for "severe"; after 2 severe → 50x for a third

- **⚠ Antepartum fetal testing is not predictive**: testing is usually reassuring right before a recurrence.
- Ruiter recommends **induction at 37 wk**; Williams: reasonable to **induce at 38 completed weeks**.

Other associations

- **Smoking: 2x** (meta-analysis, 1.6 million); **5–8x** with chronic hypertension and/or severe preeclampsia.
- **Cocaine**: 8 of 50 users had a stillbirth from abruption.
- Serum markers: abnormal **MSAFP, inhibin, PAPP-A**; 3rd-trimester **MSAFP >280 µg/L** may predict.
- **Subclinical hypothyroidism / antithyroid antibodies: 2–3x**.
- Lupus anticoagulant → **maternal floor infarction**, less so typical abruption.
- **⚠ Thrombophilias: no convincing link** to abruption (ACOG).

Diagnosis

- Classic triad: **sudden abdominal pain, vaginal bleeding, uterine tenderness**.

Finding (Hurd, 1983)	Frequency
Vaginal bleeding	78%
Uterine tenderness or back pain	66%
Nonreassuring fetal status	60%
Misdiagnosed as preterm labor	⅓

- Also frequent contractions and **persistent hypertonus**.
- Profuse external bleeding may come with only minor separation, and a **concealed abruption** may kill the fetus with no external bleeding. Either can cause **hypovolemic shock**.
- Abruption killing the fetus: blood loss often **≥½ of the pregnant blood volume** (141 women).

Consumptive coagulopathy

- **Some intravascular coagulation is almost universal**. Abruption is the **most common cause of clinically profound consumptive coagulopathy in obstetrics** (probably in all of medicine).
- Plasminogen → plasmin lyses fibrin microemboli. With fetal death, **FDPs and D-dimers are almost always ↑**, but measuring them is **not clinically useful**.

- **Fibrinogen <150 mg/dL in 1/3** of abruptions that kill the fetus. It depends on the **pre-abruption level** (higher = "protective"). Measure to guide replacement.
- Other factors ↓ → replaced empirically with **FFP** in massive transfusion protocols. **Thrombocytopenia** may accompany; **dilutional coagulopathy** from repeated transfusion adds to DIC.
- **More likely with concealed abruption; less common with partial abruption and a live fetus.** If serious coagulopathy develops, it is **usually evident by the time symptoms appear.**
- Parkland case: presented with only a **nosebleed**, dead fetus, **fibrinogen 25 mg/dL** → total abruption.

Couvelaire uterus

- **Uteroplacental apoplexy** = widespread extravasation of blood into the myometrium and beneath the serosa, seen at cesarean.
- May cause **atony** but is **NOT by itself an indication for hysterectomy.**
- Blood may also collect under the tubal serosa, in the broad ligaments, ovaries or peritoneal cavity.



Figure 43-5. Couvelaire uterus after total abruption: blood infiltrates the myometrium to the serosa. Not an indication for hysterectomy on its own.

End-organ injury

- **AKI** from delayed or incomplete treatment of hypovolemia; **prompt, vigorous blood and crystalloid usually prevents it**, even with severe DIC. Risk ↑ with **preeclampsia**.
- Most AKI is reversible without dialysis; occasionally **irreversible acute cortical necrosis**.
- Rarely **Sheehan syndrome** (pituitary failure) after severe hemorrhage.

Differential diagnosis

- Severe abruption is usually obvious; lesser degrees are a **diagnosis of exclusion**.
- **Sonography sensitivity only 24%** (149 women) → **a negative scan does not exclude abruption** (placenta and fresh clot look alike).
- **MRI is highly sensitive:** consider it if the result would change management. D-dimer may be suggestive (not adequately tested).
- Bleeding with a live fetus → look for **previa** and other causes.
- "Painful bleeding = abruption, painless = previa" is **not reliable:** labor with previa hurts; abruption may mimic labor or be **painless (especially with a posterior placenta).**

Management

- Depends on maternal and fetal condition, gestational age and the amount of hemorrhage. **Start prompt, intensive resuscitation with blood and crystalloid** in all cases.

Situation	Approach
Live, viable fetus, vaginal delivery not imminent	Emergency cesarean (most)
Dead or previable fetus	Vaginal delivery preferred
Uncertain diagnosis, live uncompromised fetus	Close observation with immediate intervention available
Preterm fetus, no compromise	Expectant management possible; Williams: tocolysis contraindicated

- ⚠ Confirm fetal heart activity by **sonography**: a scalp electrode on a dead fetus can record the **maternal heart rate**.

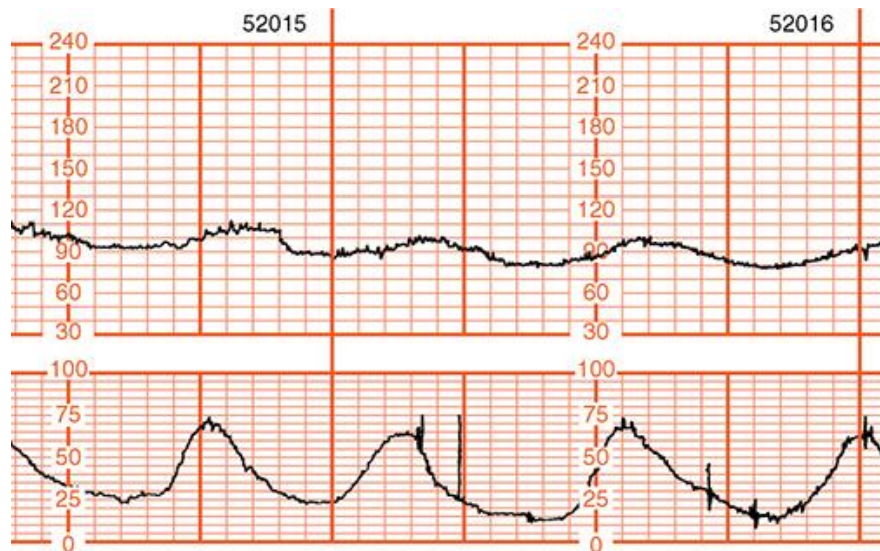


Figure 43-6. Abruptio with fetal compromise: uterine hypertonus (baseline 20–25 mm Hg, contractions to ~75 mm Hg) with fetal bradycardia and repetitive late decelerations.

Cesarean delivery

- **Speed matters.** Abruptio with bradycardia: **15 of 22 intact survivors** delivered within **20 min** of the decision; **8 of 11** who died or got cerebral palsy were delivered **after 20 min**. Decision-to-delivery interval correlates inversely with cord pH.
- Abruptio is one of the strongest antecedents of **cooling for neonatal encephalopathy**.
- Major hazard: **consumptive coagulopathy** → **two large-bore IV lines**, blood and components ready. Guide replacement with **Hb, platelets, fibrinogen, PT and PTT**.

Vaginal delivery (dead fetus)

- ⚠ **Placental-site hemostasis depends on myometrial contraction, not on coagulation** → use **uterotonics and uterine massage**.
- Cesarean may still be needed for **brisk hemorrhage** not controlled by vigorous replacement, or other obstetrical indications.
- Labor may be rapid because of **persistent hypertonus**: baseline intraamniotic pressure **≥50 mm Hg**, contractions **>100 mm Hg**. Overall 1st and 2nd stages are **not shorter** than usual.
- **Early amniotomy**: long championed (spiral artery compression, less thromboplastin entry); no proof, but may hasten delivery. An intact sac may dilate the cervix better if the fetus is small.
- **Oxytocin in standard doses** if rhythmic contractions are not superimposed on hypertonus; it does **not** worsen coagulopathy.
- **Misoprostol** is less favored (**tachysystole**).
- ⚠ **No arbitrary time limit** to delivery: maternal outcome depends on **diligent fluid and blood replacement**, not the interval. Women transfused for **≥18 h** before delivery did as well as those delivered sooner.

Expectant management (immature fetus)

- 43 women <35 wk (31 given tocolysis): delivery gained **~12 days**; cesarean **75%**; no stillbirths.
- CAOS: abruptio at ~20 wk → oligohydramnios → delivery ~28 wk.

- Abrupton <28 wk (256 women): mean gain **1.6 wk**; **65%** delivered <29 wk; half emergent cesarean.
- ⚠️ **Continuous FHR monitoring does not guarantee safety**: a normal tracing may precede sudden further separation and fetal death.
- **Tocolysis**: PMR **5% with or without** it (131 women <36 wk); one RCT (MgSO₄ at 24–34 wk) showed no benefit. **Terbutaline tachycardia may mask maternal compromise.**
- **Williams**: suspected abrupton contraindicates tocolytics.

2. Placenta Previa

Classification (NIH Fetal Imaging Workshop, 2014)

- *Previa* = "going before": the placenta precedes the fetus into the birth canal, implanted in the **lower uterine segment** over or very near the internal os.

Term	Definition
Placenta previa	Internal os covered partially or completely by placenta (formerly "total" or "partial" previa)
Low-lying placenta	Edge does not cover the os but lies within 2 cm of it (formerly includes "marginal previa")
Vasa previa	Fetal vessels run through the membranes over the os (see Ch 6)

- Classification can **change with cervical dilation** (low-lying at 2 cm may become partial previa at 4 cm; total may become partial).
- ⚠️ **Digital palpation to assess the placental edge usually causes severe hemorrhage!**
- Some separation (bleeding) is **inevitable** as the lower segment forms and the cervix dilates; this is not termed abrupton.

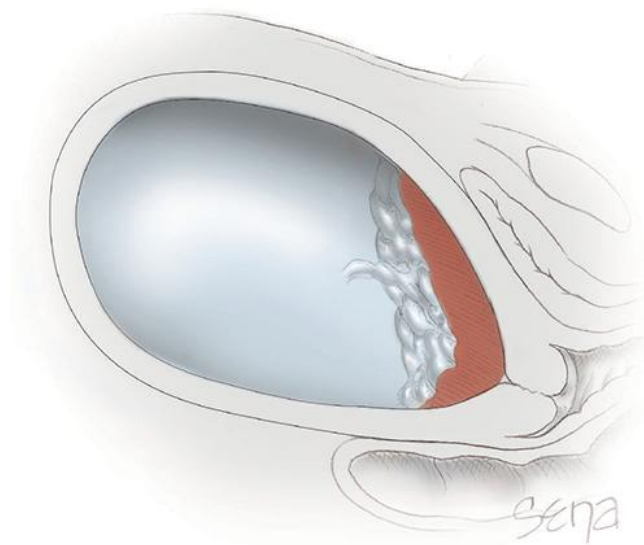


Figure 43-7. Placenta previa covering the internal os: any cervical dilation will cause copious bleeding.

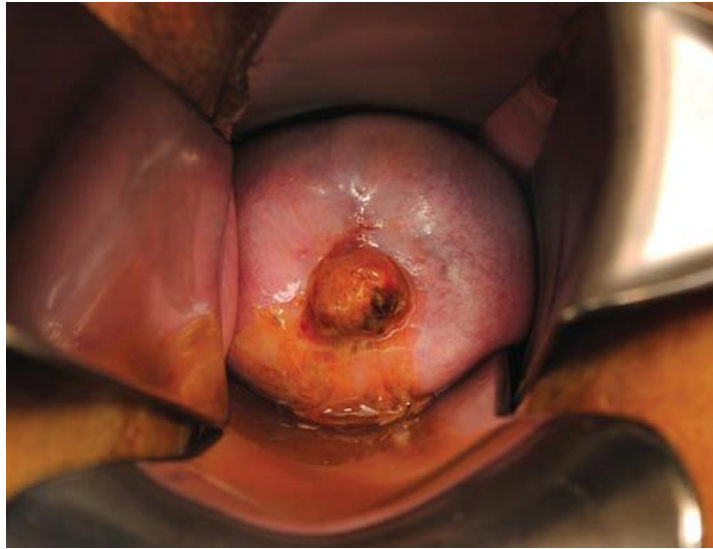


Figure 43-8. Speculum view: placenta protruding through the cervical os.

Placental "migration"

- A **misnomer**: villi are anchored and do not move. Apparent movement is from:
 1. Imprecise **2-D sonography**
 2. **Differential growth** of the lower vs upper segments
 3. **Trophotropism**: growth toward the better-perfused fundus
- Many "migrating" placentas never truly covered the os circumferentially.
- **Less likely to migrate with a prior cesarean scar** (previa more likely to persist).
- Midpregnancy (4300 women): **12% low-lying**. If **not covering** the os → previa did not persist, no bleeding. If **covering** the os → **~40% persisted** to delivery.
- A placenta near but not over the os until the early 3rd trimester is **unlikely to persist**.

Incidence and risk factors

- **~0.4% (1 in 250–400)** deliveries; rising. Parkland: **2.6/1000** (1988–2003) → **3.8/1000** (2004–2020).

Factor	Data
Maternal age	FASTER: 0.5% if <35 vs 1.1% if ≥35. Parkland: ~0.65/1000 at ≤19 yr → ~10/1000 at >35 yr
Multiparity	2.2% with parity ≥5 (age and parity confounded); interpregnancy interval no effect
Smoking	≥2× (CO hypoxemia → placental hypertrophy; decidual vasculopathy)
Prior cesarean (foremost)	1.3% with 1 prior → 3.4% with ≥6 prior (MFMU, 30,132). Cesarean in 1st pregnancy → 1.6× previa in 2nd. Parity >4 with >4 cesareans → 8×
Multifetal gestation	↑; dichorionic > monochorionic (two implantation sites)
Unexplained ↑ MSAFP	↑ previa; previa with MSAFP >2.0 MoM at 16 wk → more late bleeding and preterm birth
ART	↑ even after adjusting for multiples
Leiomyomas	↑ (systematic review, ~256,000 births)

- ⚠ **Previa + prior uterine incision → PAS and hysterectomy**: hysterectomy in **6%** with previa at primary cesarean vs **25%** at repeat cesarean.

Clinical features

- **Painless bleeding** is the most characteristic event, usually **near the end of the 2nd trimester or later** (can be before midpregnancy; some late abortions).
- The first **"sentinel bleed"** comes without warning, is **rarely fatal**, usually stops, then **recurs**.

- **~10%** (especially placenta near, not over, the os) do not bleed **until labor**; this can mimic abruption.
- **Mechanism:** lower segment forms and the os dilates → placental separation. Bleeding is worse because **lower-segment myometrium contracts poorly** and cannot constrict torn vessels. It also continues **after placental delivery**; friable cervix and lower segment lacerate (especially after manual removal).
- **Cervical length:** 3rd-trimester **<30 mm** → more hemorrhage, contractions and preterm birth (one study). **<25 mm** at admission for bleeding → more delivery within 7 days (one study). Other studies disagree.
- **PAS is a frequent, serious association:** poorly developed lower-segment decidua. Placental bed biopsies: spiral arterioles infiltrated by **trophoblastic giant cells in 50%** vs 20% of controls. Abnormal attachment in **7%** of 514 previas.
- **⚠️ Coagulopathy is rare** with previa, even with extensive separation: thromboplastin escapes through the cervix, and the area has **few large myometrial veins**.

Diagnosis

- Bleeding after midpregnancy → always consider **previa and abruption**. Previa = **21%** of admissions for bleeding at **22–28 wk**.
- **⚠️ Do not exclude previa until sonography clearly proves its absence.**
- No sonography available → digital exam only in the **OR with a "double set-up"** (prepared for immediate cesarean). Even the gentlest exam can cause torrential bleeding.
- **Transabdominal** sonography first: excellent sensitivity and NPV when the placenta clearly overlies the cervix or lies away from the lower segment.
 - Pitfalls: **obesity**; a **full bladder** elongates the cervix and compresses the lower segment, **mimicking previa**.
- **Transvaginal sonography = most accurate**, and **safe even with bleeding**. Internal os seen in **100%** transvaginally vs **30%** transabdominally.
- **MRI:** excellent images but won't replace ultrasound for routine use; **useful for PAS**.



Figure 43-9. Transvaginal sonogram at 21 wk: posterior placenta (P) completely covers the internal os (arrowhead); dotted line = cervical canal.

Parkland follow-up imaging

Finding at 18–22 wk	Re-image
Previa + prior cesarean	28 wk , then 32 wk if persistent
Previa, no prior cesarean	32 wk
Edge still <2 cm from os at 32 wk	TVS again at 36 wk

- Activity restriction only if previa persists **beyond 28 wk** or with bleeding/contractions earlier.

Management

- Individualized by **fetal maturity, labor and bleeding severity**. Emergency delivery in **43%** (half preterm) of 214 women; overall **25–40%**.
- **Bleeding subsides, fetus immature** → close observation in an obstetrical unit.
- **Tocolytics**: some limit to **48 h**; ⚠ Williams **categorically recommends against** them (hypotension/tachycardia mask maternal compromise).
- **Discharge** after bleeding has stopped for **~2 days** with a healthy fetus → "**pelvic rest**", must be able to return immediately. **SMFM: no routine cervical length screening**.
- Long-term inpatient care adds **no benefit** over outpatient care in selected women. RCT (53 women, 24–36 wk): no difference; **60% rebled**, **half** needed expeditious cesarean.
- **Timing (ACOG/SMFM): uncomplicated previa** → **deliver at 36⁰/₇–37⁶/₇ wk**. Parkland: elective cesarean usually at **38⁰/₇ wk**.

Delivery

- **Practically all by cesarean**. Some prefer a **vertical skin incision** (fast entry, room for hysterectomy).
- **Emergent in >1/2** for hemorrhage; **~1/4** need transfusion.
- Low transverse hysterotomy is usually possible, but cutting an **anterior placenta** may cause **fetal bleeding** → deliver quickly; a **vertical uterine incision** may be better at times. Outcomes are rarely compromised even when incising through the placenta.

Controlling implantation-site bleeding (poorly contracting lower segment)

1. **Uterotonics and pressure**
 2. **Oversew** the site with **no. 0 chromic**; **Cho sutures** (interrupted sutures at **1-cm intervals** in a circle around the bleeding area); **compression sutures** through anterior and posterior walls
 3. **Bakri or Foley balloon tamponade** ± compression sutures
 4. **Bilateral uterine or internal iliac artery ligation**; **pelvic artery embolization**
 5. **Hysterectomy** if these fail and bleeding is brisk
- Prophylactic **internal iliac balloon occlusion** did **not** reduce PPH or morbidity (RCT).
 - Hysterectomy rate with previa **without PAS ~2%**. Peripartum hysterectomy: **40%** done for abnormal placentation (UK, 318); **44%** of Parkland cesarean hysterectomies were for bleeding previa or PAS.

Maternal and perinatal outcomes

- **Maternal mortality ~3×**; previa alone caused **~3%** of US hemorrhage deaths (2006–2013).
- **Preterm delivery** is the main cause of perinatal death; neonatal death is ↑ even **at term**.
- **Fetal anomalies 2–3×** more common.
- **FGR**: association minimal after controlling for gestational age (most LBW is preterm); slight ↑ in a meta-analysis. Similar with PAS.

3. Placenta Accreta Spectrum (PAS)

- **PAS** = abnormally implanted, invasive or adhered placenta (Latin *ac-* + *crescere*, "to adhere"). Also called the **morbidity adherent placenta**; in Europe, "pernicious placenta previa with accreta."

Etiopathogenesis

- Main problem: the placenta **fails to separate** from the myometrium after delivery.
- Partial or total **absence of the decidua basalis** and imperfect **Nitabuch (fibrinoid) layer** → no physiological cleavage plane → **villi attach directly to smooth muscle**.

- Not just decidual deficiency: **cytotrophoblast-controlled invasion** (angiogenesis), immunologically mediated "**hyperinvasiveness**", and highly expressed remodeling/adherence genes.
- **Myometrial fibers on the basal plate** in a prior pregnancy predict later PAS → a "**constitutional endometrial defect**".
- Prior surgical trauma increases vulnerability to invasion. **Cesarean scar pregnancy (CSP) is a precursor**: CSP and PAS lie on one spectrum. Early rupture and hemorrhage are common with CSP, so many women choose termination.

Classification (by depth of invasion)

Variant	Villi	Approximate frequency (general)	Referral center
Accreta	Attached to the myometrium	80%	22%
Increta	Invade the myometrium	15%	24%
Percreta	Penetrate through the myometrium to or through the serosa	5%	54%

- **Total** = all lobules abnormally adherent; **focal** = all or part of a single lobule.
- Histological diagnosis needs **myometrium** (placenta alone is not enough). FIGO has proposed a clinical classification.

Memory hook: Accreta **A**ttaches, Increta **I**nvasives, Percreta **P**enetrates (80 : 15 : 5).

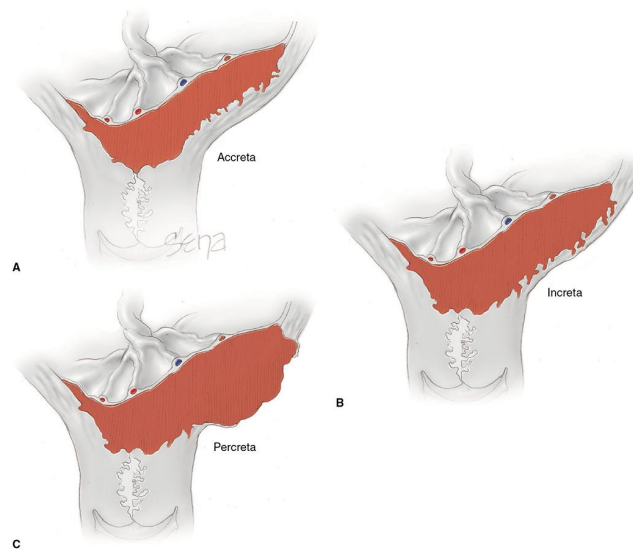


Figure 43-10. PAS: (A) accreta, (B) increta, (C) percreta through the serosa toward the bladder.

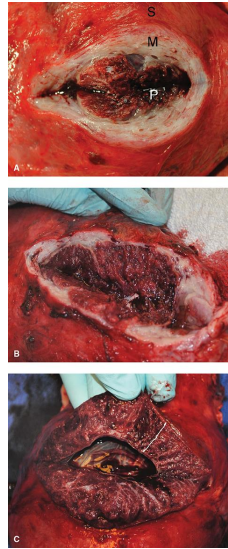


Figure 43-11. Increasing invasion at surgery: (A) minimal, myometrium (M) intact; (B) deeper; (C) placenta reaching the serosa with no myometrium left.

Incidence and risk factors

- **1 in 20,000** (1951) → **1 in 2500** (1980s) → **1 in 731** (MFMU, 115,502 women) → **1 in 270** (Nationwide Inpatient Sample). Rise is **directly related to the cesarean rate**.
- **13%** of US pregnancy-related deaths (2006–2013) were from PAS hemorrhage. A leading cause of **emergency peripartum hysterectomy**.
- **Recurrence ~20%** if the uterus is conserved; also ↑ previa, uterine rupture, manual removal, preterm delivery, hysterectomy.

Risk factors (similar to previa)

- **The two most important: placenta previa and prior cesarean delivery, especially both.**
- **Classical** hysterotomy > low-segment scar.
- **Almost half** of women with a prior cesarean have myometrial fibers adherent to the placenta microscopically.
- Other myometrial trauma: **curettage, endometrial ablation. ART** is an independent risk factor.
- **MSAFP >2.5 MoM → 8x; free β-hCG >2.5 MoM → 4x** (>9300 women screened at 14–22 wk).

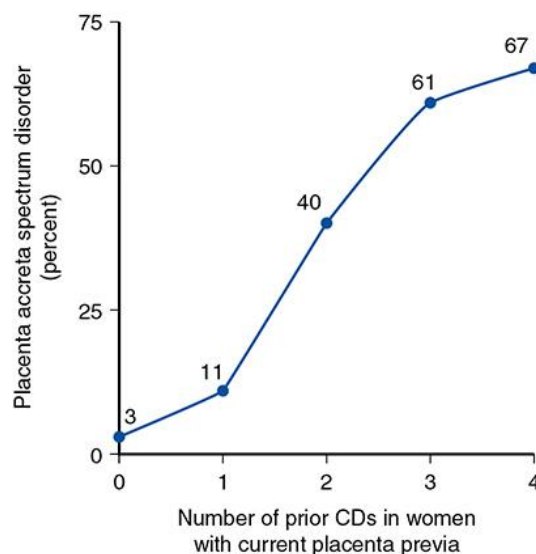


Figure 43-12. PAS risk with a current previa: 3% (no prior cesarean) → 11% (1) → 40% (2) → 61% (3) → 67% (≥4 prior cesareans).

Diagnosis

- 1st- and 2nd-trimester bleeding with PAS usually comes from a **coexisting previa** → sonography.

- 1st trimester: the **smallest myometrial thickness** helps predict later hysterectomy. Screening at 11–13 wk (>22,000): **6%** high risk; 14 suspected, 13 confirmed at delivery (imaging still imperfect).

Five sonographic signs of PAS

1. **Placental lacunae** ("lakes")
2. **Thinning of the retroplacental myometrium**
3. **Disruption of the bladder–uterine serosal interface**
4. **Bridging vessels** from placenta to the bladder–serosal interface
5. **Placental bulge** distorting the uterine contour

Performance	Accreta	Increta	Percreta
Sensitivity (meta-analysis)	91%	93%	89%
Specificity	97%	98%	99%

- ⚠ These are likely **overestimates** (clinical bias, interobserver variability). Bowman: sensitivity **54%**, specificity **88%**, PPV **82%**, NPV **65%**, accuracy **65%**.
- **Location matters:** detection **90% anterior vs 50% posterior**; posterior PAS → delayed diagnosis and more surgical complications.
- Interpret sonography **with clinical and operative findings**.
- **Color/power Doppler and 3-D** help: invasion suspected if **serosa–bladder interface to retroplacental vessels <1 mm** plus **large lacunae**. **Hypervascularity of the serosa–bladder interface** has the best PPV and NPV for **percreta**. **Intracervical lakes** also reported.
- **Placenta accreta index:** number of prior cesareans, placental location, lacunae, bridging vessels, smallest myometrial distance.



Figure 43-13. Transabdominal sonogram of percreta: multiple large placental lacunae (L).

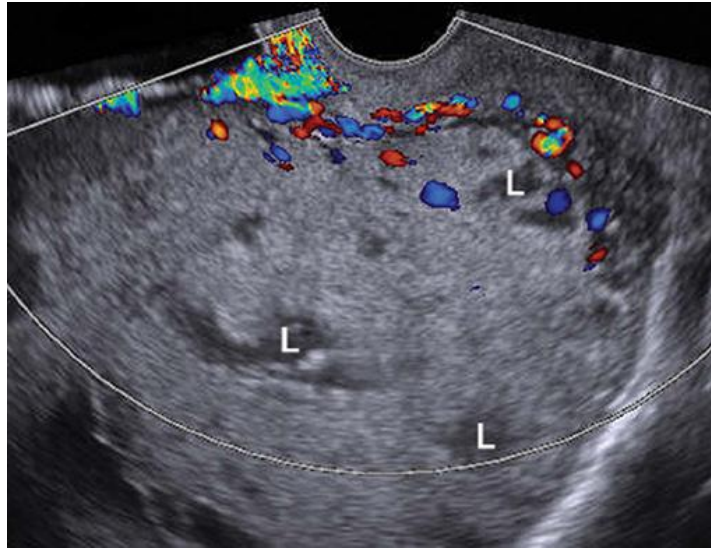


Figure 43-14. Transvaginal color Doppler in percreta: retroplacental vessels invading the myometrium, with intraplacental venous lakes (L).

MRI

- Defines **invasion of adjacent structures**; good interobserver agreement for diagnosis and depth (less for topography). Shows **pelvic parasitic arteries**. Not all find it helpful.
- **Gadolinium usually not used** in pregnancy (may enhance images).
- **Three MRI signs (Lax): uterine bulging, dark intraplacental bands on T2, heterogeneous placental signal** (lacunae).
- Use when sonography is **inconclusive** or with a **posterior previa**.

Management: planning

- Key decisions: **timing and facility**. Needs surgical, anesthesia, ICU and **blood bank** capability, plus an obstetrical surgeon or **gynecological oncologist** and surgical, urological and **interventional radiology** consultants.
- **ACOG/SMFM: planned delivery in a tertiary-care facility**. Dedicated on-call teams improve maternal outcomes (meta-analysis).
- Schedule for peak resource availability, but **1/3 need unscheduled delivery** → have contingency plans. Women who **refuse blood** are especially difficult.

Table 43-2. Criteria to Consider for Delivery of Placenta Accreta Spectrum (PAS) in a Center of Excellence

Criteria
Sonogram findings suspicious for PAS
Placenta previa with abnormal ultrasound appearance
Placenta previa with 2 to 3 prior cesarean deliveries
Prior classical cesarean delivery and anterior placentation
Prior endometrial ablation or pelvic irradiation
Inability to adequately evaluate or exclude PAS
Any other reason to suspect PAS

Delivery timing

- **40%** bled and needed **unplanned delivery before 34 wk** (one study).

Source	Recommended timing
ACOG	Individualize; 34⁰/₇–35⁶/₇ wk (no lung maturity testing needed after 34 wk)
SMFM	34–37 wk
Surveys of practice	Most deliver at ≥36 wk
Parkland	Generally after 36 completed weeks , ready for nonelective cases

- **⚠ PAS found unexpectedly at laparotomy** with inadequate resources, and the woman is stable and not bleeding → **do not deliver the fetus**; close the abdomen and **transfer** to a tertiary center.

Prophylactic catheters

- **Ureteral catheters**: may aid identification and dissection but **do not lower urinary tract injury** overall; used by some.
- **Arterial balloon catheters** (internal iliac, or infrarenal aortic), inflated after fetal delivery, or used to embolize: **borderline efficacy, serious risks** (e.g., common iliac thrombosis). **ACOG: no recommendation for or against.** Prophylactic **internal iliac ligation: no benefit.**

Cesarean delivery and hysterectomy

- Estimate the need for hysterectomy beforehand. Partial PAS may allow placental delivery with **hemostatic sutures**. **⚠ Increta or percreta almost always → hysterectomy.**
- **Steps:**
 1. Before hysterotomy, dissect early: **wide bladder flap**, divide the **round ligaments**, carry the peritoneal incisions down around the implantation site (prevesical space, posterior bladder wall).
 2. **Hysterotomy away from the placenta: classical, transverse fundal or high posterior.**
 3. After the fetus is out, assess the invasion **without attempting manual placental removal**. Attempted partial or total removal before hysterectomy **doubles blood loss**.
 4. Obvious increta/percreta → **hysterectomy with the placenta left in situ**. Bleeding may be minimal until removal is tried.
- Heavy bleeding: immediate blood replacement plus **uterine or internal iliac artery ligation, balloon occlusion or embolization**. **Supracervical hysterectomy** may be easier. Kingdom **modified radical hysterectomy** (Table 43-3).
- **High postoperative VTE risk → consider thromboprophylaxis.**

Table 43-3. Steps in Cesarean Hysterectomy for Management of Placenta Percreta

Step	Details
Midline vertical laparotomy and midline hysterotomy	Enter uterus to avoid placenta
Superior devascularization	Divide and ligate uteroovarian pedicles and round ligaments
Retroperitoneal dissection	Incise down to paravesical space; consider cephalad dissection to expose ureters and bifurcation of common iliac arteries
Bladder dissection	Progress lateral to medial and down to vaginal fornices; associated with troublesome venous bleeding
Colpotomy	Inferior dissection of extremely vascular paravesical space may cause significant hemorrhage

Conservative management (placenta left in situ)

- Trim the cord, close the hysterotomy, leave the placenta, **no hysterectomy**. Options for:
 - **Unsuspected PAS** at cesarean when uterine closure stops bleeding → **transfer** for definitive care.

- Women strongly desiring **fertility** after extensive counseling. Resecting the adherent placenta with its myometrium is a uterine-sparing alternative; external uterine sutures over the placental bed (**Naisicaa** procedure) are another.
- The placenta resorbed spontaneously in **1–12 months (mean 6)**.
- Complications: **sepsis, coagulopathy, pulmonary embolism, AV malformation**.
- Delayed hysterectomy (planned, or for bleeding or infection) days to weeks later: needed in **≤20%** in some studies but **up to 60%** in others.
- **Methotrexate: no evidence** it helps. **Serial β -hCG is uninformative** → follow with serial **sonography or MRI**.
- **ACOG (and Williams): leaving the placenta in situ is investigational**, except to allow transfer. ISAIP guidelines are more permissive.

Pregnancy outcomes

- Invasion depth does **not** predict perinatal outcome but is of **paramount maternal significance**.
- Even with prenatal diagnosis at tertiary centers: hemorrhage, **urinary tract injury**, ICU admission, reoperation and maternal death. **Worse** without tertiary care or when percreta is found only at delivery.

Table 43-4. Selected Maternal Outcomes in Women with Placenta Accreta Spectrum Identified Prenatally and Delivered in Tertiary-Care Units

Outcome ^a	San Diego ^b n = 62	Utah ^c n = 60	Houston ^d n = 189	Adelaide ^e n = 67	Montreal ^f n = 42
Gestational age (wk)	33.0 ± 1.1	34 (17–41)	34 (32, 35)	35 (32, 36)	NS
Operating time (min)	194 ± 1.6	NS	NS	NS	112 ± 49
Transfusions	~75%	70%	NS	43%	34%
RBC (units)	4.7 ± 2.2	≥4 (30%)	2 (0–5)	4 (3, 6)	4
Bladder injury	23%	37%	3.7%	6%	13%
ICU admission	72%	30%	100%	NS	9%
LOS (days)	7.4 ± 1.8	3–13	NS	7.9	5.2

^aOutcomes shown as mean ± 1 SD; median (range). ^bData from Warshak, 2010. ^cData from Eller, 2011. ^dData from Erfani, 2019. ^eData from Yasin, 2019. ^fData from Mitric, 2019. ICU = intensive care unit; LOS = length of stay; NS = not stated; RBC = red blood cells.

Quick-Recall: Abruptio vs Previa vs PAS

Feature	Placental abruptio	Placenta previa	Placenta accreta spectrum
Definition	Separation of a normally sited placenta before delivery	Placenta covers the internal os (low-lying = within 2 cm)	Villi attach to / invade / penetrate myometrium (absent decidua basalis)
Incidence	~1 in 200 (Parkland 1 in 290)	~1 in 250–400 (Parkland 3.8/1000)	1 in 270–731 and rising
Key risk factors	Prior abruptio , hypertension, PPROM, smoking, cocaine, trauma	Prior cesarean , age, parity, smoking, multifetal, ART	Previa + prior cesarean , classical scar, ablation, ART, ↑ MSAFP
Bleeding	Painful , may be concealed ; hypertonus	Painless , recurrent "sentinel" bleed	From coexisting previa; massive at attempted placental removal
Coagulopathy	Common (most frequent obstetric DIC)	Rare	From massive hemorrhage
Sonography	Insensitive (24%) ; MRI highly sensitive	TVS is accurate and safe	5 signs (lacunae etc.); poorer for posterior ; MRI adjunct
Delivery	Live viable fetus → cesarean ; dead fetus → vaginal	Cesarean at 36⁰/₇–37⁶/₇ wk (Parkland 38)	Planned cesarean hysterectomy , tertiary center, 34⁰/₇–35⁶/₇ wk (Parkland >36)
Don't	Use tocolytics ; set an arbitrary delivery time limit	Do a digital exam (outside a double set-up); use tocolytics	Try manual placental removal ; incise through the placenta

Key Numbers to Memorize

- Intervillous flow at term **>600 mL/min**
- Abruptio **0.5% (1/200)**; fetal-death abruptio now **1 in 2060** (was 1 in 420); **>50%** at ≥37 wk
- Abruptio PMR **119/1000** vs 8/1000; **10–12%** of 3rd-trimester stillbirths; **15%** of liveborns die
- Fetomaternal hemorrhage in spontaneous abruptio **20%**, **all <10 mL**; traumatic abruptio **80–100 mL**
- Recurrence **5.8–22%**; **12x** after severe; **50x** after two severe; induce at **37–38 wk**
- Presentation: bleeding **78%**, tenderness/back pain **66%**, fetal compromise **60%**, "preterm labor" **1/5**
- Fibrinogen **<150 mg/dL** in **1/3** of fetal-death abruptions; ultrasound sensitivity **24%**
- Decision-to-delivery **≤20 min** for bradycardia; intraamniotic baseline **≥50**, contractions **>100 mm Hg**
- Previa **0.4%**; low-lying **<2 cm**; 12% low-lying at midpregnancy, **~40%** of os-covering previas persist
- Previa with prior cesarean: hysterectomy **25%** (vs 6%); previa **1.3% → 3.4%** with 1 → ≥6 cesareans
- Previa: **~10%** first bleed in labor; PAS in **7%**; hysterectomy without PAS **~2%**; maternal mortality **3x**
- Previa delivery **36⁰/₇–37⁶/₇ wk**; discharge after **~2 days** without bleeding
- PAS accreta : increta : percreta = **80 : 15 : 5**; PAS with previa **3 / 11 / 40 / 61 / 67%** for 0–4 prior cesareans
- PAS **13%** of US hemorrhage deaths; recurrence **~20%**; MSAFP >2.5 MoM **8x**
- PAS US sensitivity **~90%** (meta-analysis) but **54%** in practice; anterior **90%** vs posterior **50%**; serosa–vessel distance **<1 mm**
- PAS delivery **34⁰/₇–35⁶/₇ wk** (ACOG); **40%** bleed before 34 wk; **1/3** unscheduled
- In-situ placenta resorbs in **1–12 mo (mean 6)**; later hysterectomy **≤20% to 60%**